INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Google: Arcola 2, VA -- station KIAD, America/New_York
Committed pair	OSM 693381107 -> 545396372 Amazon IAD81 / Amazon IAD62
Facade gap	137.7 m between the two halls
Weather record	43,763 real hourly records from KIAD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3539 C at 225 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-03-11 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.329	18.0	4.132	1.053
01	mechanical	yes	switch budget	4.329	18.0	4.132	0.994
02	mechanical	yes	switch budget	4.049	18.0	3.022	1.163
03	mechanical	yes	switch budget	2.918	18.0	1.183	1.018
04	mechanical	yes	switch budget	2.944	18.0	0.802	0.992
05	mechanical	yes	switch budget	2.359	18.0	0.242	1.168
06	mechanical	yes	switch budget	0.999	18.0	-0.868	0.983
07	mechanical	yes	switch budget	2.664	18.0	0.242	1.636
08	mechanical	yes	switch budget	6.231	18.0	5.579	1.912
09	mechanical	yes	switch budget	9.974	18.0	11.442	2.358

10	mechanical	yes	switch budget	11.931	18.0	14.216	2.254
11	mechanical	yes	switch budget	15.598	18.0	17.799	1.798
12	mechanical	no	dry-bulb	18.603	18.0	19.564	1.367
13	mechanical	no	dry-bulb	19.906	18.0	21.364	1.173
14	mechanical	no	dry-bulb	22.220	18.0	22.023	1.293
15	mechanical	no	dry-bulb	22.629	18.0	23.103	1.070
16	mechanical	no	dry-bulb	23.301	18.0	23.103	1.143
17	mechanical	no	dry-bulb	22.874	18.0	22.543	1.416
18	mechanical	no	dry-bulb	22.479	18.0	21.219	1.328
19	mechanical	no	dry-bulb	20.946	18.0	19.014	1.418
20	mechanical	no	dry-bulb	20.161	18.0	17.799	1.558
21	mechanical	no	dry-bulb	18.825	18.0	15.859	1.691
22	FREE	yes	-	16.065	18.0	12.473	1.374
23	FREE	yes	-	15.815	18.0	13.362	1.137

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.329 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.329 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.049 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.918 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.944 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.359 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.999 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.664 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.231 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.974 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.931 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.598 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.603 C against a 18.0 C limit, so it fails by 0.603 C. It would take a limit 0.603 C higher, or a bound 0.603 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.906 C against a 18.0 C limit, so it fails by 1.906 C. It would take a limit 1.906 C higher, or a bound 1.906 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 18.0 C limit, so it fails by 4.220 C. It would take a limit 4.220 C higher, or a bound 4.220 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.629 C against a 18.0 C limit, so it fails by 4.629 C. It would take a limit 4.629 C higher, or a bound 4.629 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.301 C against a 18.0 C limit, so it fails by 5.301 C. It would take a limit 5.301 C higher, or a bound 5.301 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.874 C against a 18.0 C limit, so it fails by 4.874 C. It would take a limit 4.874 C higher, or a bound 4.874 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.479 C against a 18.0 C limit, so it fails by 4.479 C. It would take a limit 4.479 C higher, or a bound 4.479 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.946 C against a 18.0 C limit, so it fails by 2.946 C. It would take a limit 2.946 C higher, or a bound 2.946 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.161 C against a 18.0 C limit, so it fails by 2.161 C. It would take a limit 2.161 C higher, or a bound 2.161 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.825 C against a 18.0 C limit, so it fails by 0.825 C. It would take a limit 0.825 C higher, or a bound 0.825 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.065 C, which is 1.935 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.374 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.1849 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.815 C, which is 2.185 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.137 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.1841 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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